

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0878

Silver Side Road 138 kV
June 02, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0878. The Interconnection Customer proposes a 124.9 MW solar photovoltaic facility interconnecting at the Silver Side Road 138 kV substation in the Delaware area. The request was received on October 14, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 312 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$853,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Lake Catherine – Bridgeport 138.0 kV	Chilliwick – 26Th And F 138.0 kV L	104.4	462	7.1
Castleton – Plainfield 138.0 kV	Bridgeport – Marcus Hook Energy Lp 138.0 kV	100.7	291	8.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bridgeport – Lake Catherine 138.0 kV Line	Reconductor line	\$639,000	20% (\$128,000)
Castleton – Plainfield 138.0 kV	Reconductor line	\$1,993,000	36% (\$725,000)
	Total allocated to project		\$853,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.